

Ziyan Pan (Pansy) - 40252218

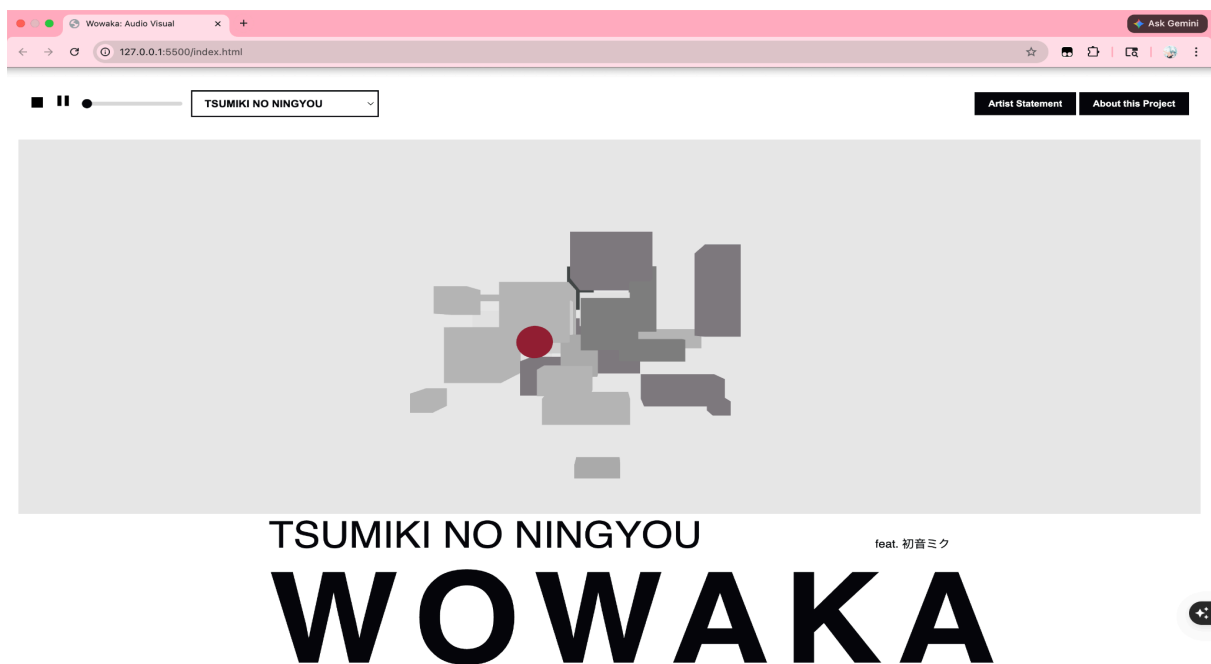
Ben Woschitz - 40286288

CART 263-B

April 16th, 2026

Honoring the legacy of Wowaka

This project acts as an audio-visual tribute to the late Wowaka, who went on to inspire millions with his music.



For our final project, we decided to continue developing our midterm project by integrating Three.js into it. This was quite challenging because our midterm concept was based entirely on 2D illustrations. Since wowaka's song cover art is also very minimalistic, it was difficult to use 3D modeling while still preserving the simplicity of a flat visual style. For our model, we chose not to use reflective textures or strong lighting, which reduced the visibility of the model's edges. By eliminating differences in color and shading between the surfaces, we were able to make the 3D model appear much flatter and more two-dimensional, which better recreated the style of the original song cover.

“裏表ラバーズ (里表情人).” Uploaded by R-A-L, *Bilibili*, www.bilibili.com/video/BV1rS4y1j7JF/. Accessed 12 Apr. 2026.



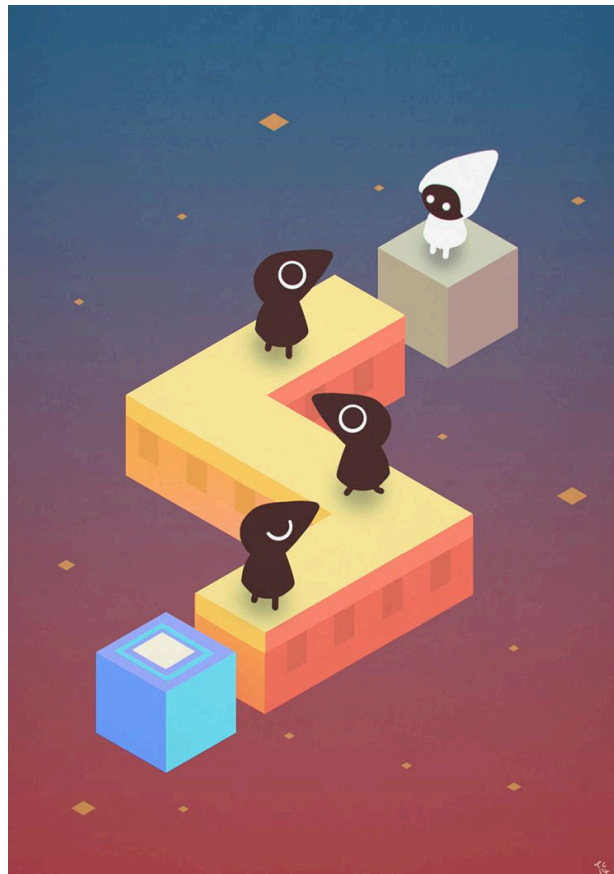
For inspiration, we looked at a fan-made PV of another Wowaka song, *Ura-Omote Lovers*. In this video, the creator was able to use very minimal 3D block models while still achieving highly dynamic visual effects. As seen in the screenshot above, the models were rendered without reflective surfaces, which softened the edges and allowed the visuals to maintain a more 2D appearance. The uploader also incorporated extensive stretching and reshaping of the models, along with the use of ease-in and ease-out animations, which we found particularly inspiring. From this work, we learned several techniques for making animations feel more synchronized with the rhythm of the music.

We also faced another challenge: Pansy was not very experienced with 3D modeling, while Ben was much more skilled in that area. To reduce Ben’s workload, since he was responsible for modeling two of the songs, Pansy chose a song cover that consisted mainly of simple rectangles and circles. The song was *Sumiki no Ningyou*, which translates to *Doll of Building Blocks*. Because the reference image was not overly complex, Pansy was able to construct some of the blocks directly through code and animate them based on the pitch of the song. Ben was then able to create a more polished and complex model in Blender. Pansy used it as a background element and applied slight scaling changes based on the pitch of the music. Through this approach, we were able to make the entire piece more dynamic and visually engaging, while using less code and fewer animations.

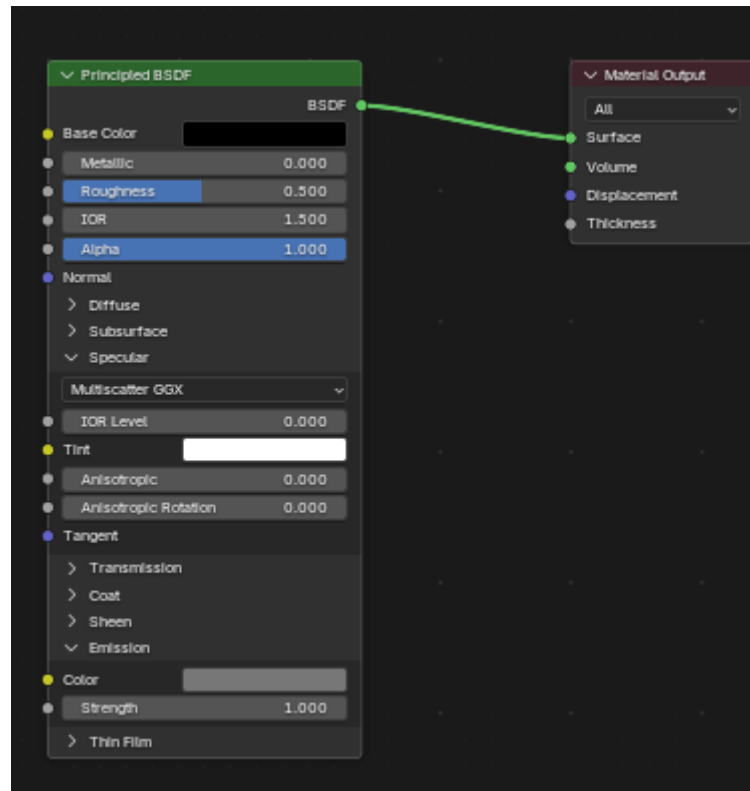
Ben’s model also significantly reduced Pansy’s workload, as it eliminated the need to manually construct each block through code. In addition, it improved the performance of the

webpage. The previous manually coded version often lagged or failed to render properly, whereas Ben's model provided a much smoother and more stable experience.

I (Ben) will just be talking in first person about the process I had for our project. First off, I chose the song Tosenbo (off limits) as the one I wanted to work on. As I was browsing the song selection, it wasn't about choosing what was easiest to model, but rather, I looked for a song with 2D visuals that I can easily create effects for. A lot of Wowaka's song covers are flat and 2D images represented by simple shapes. This is great for 2D, but as we're working with 3D, I needed something that I could work with. This led to me choosing Tosenbo. Both Ningyou and Tosenbo were very easy models to recreate from 2D to 3D as they are just made up of mainly cubes and rectangles. I was also inspired by and referenced a game I have played before named Monument Valley, which uses very simple and flat shading as seen here.



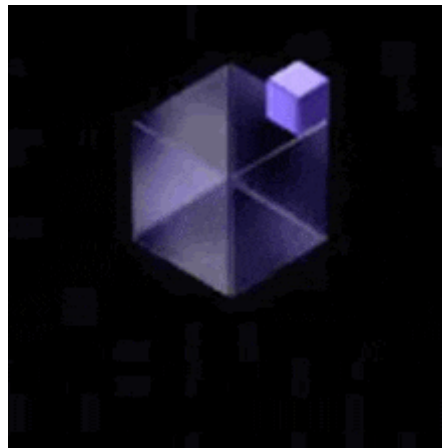
I learned how to somewhat recreate this art style for my game in my CART 315 class, and decided to use part of that method to give the cubes and rectangles the flat aesthetic that you see in our project. I then exported the 2 blender files as .glb to keep the materials intact and were easy to implement within our project. I have worked with [Three.js](#) a bit before so I was already familiar as to how 3D models work in this context.



The process for the shading was quite simple, only requiring that base colour be set to black, turning off specular and setting emission strength to 1 with the chosen colour. It's the small things like this that really make a big difference, as if our models had any kind of shading or specular lighting, it would have felt off with the rest of the project and aesthetic.



As with our other songs, I did not want to completely change the visual style for the cover, and I wanted to stay true to what we have. Tosenbo is a song about isolation and wanting to withdraw from society, and these different cubes I interpreted as chaotic everyday life, with off limits keeping the viewer from engaging in it, therefore isolating themselves. I decided to amplify this theme even further two different ways. For one, I wanted to take advantage that we were working with 3D models and so I decided to have it rotate on itself at set intervals on its X and Z axis. Coupled with the increased scale of the model, the viewer is essentially inside of it, being overwhelmed by many different rotating and moving cubes that follow the song's rhythm through amplitude. I then decided to take the duration of the song and push this narrative of isolation by having the model slowly move away from the viewer over the course of the song; therefore escaping chaotic everyday life. I was also inspired to give the model a rotation similar to the gamecube logo cube that can be seen moving around on its different faces. Rather than a constant rapid rotation, I slowed it down using a specific interval to keep a sense of normalcy across all of the chaos.



PLEASE NOTE

We had issues with Github after completing the project. Despite adding and modifying onto our previous midterm, Github decided to split our old work and new work into 2 separate files, with the old work in cart263-final and our new files in cart263-Final (capital F). We have no idea why, and we got external help, but still had issues. Therefore our link did not initially work. This was not reflected on either of our ends because neither of us did this on our computers, and we had no issues running everything locally as everything was in one place on both of our ends. Our bandaid solution was to duplicate and create a new folder with the suffix of "FinalFinal." So, our commits may appear strange and a bit all over the place.